



KLE Technological University

Creating Value,
Leveraging Knowledge

Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

Machine Learning Report
on
Product Recommendation System

By:

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Under the Guidance of

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CERTIFICATE

This is to certify that project entitled “**Product Recommendation System**” is a bonafide work carried out by the student team of “**Nandeesh katkol-02FE22BEC040, Priyanshu Porwal-02FE22BEC056, Jamuna Pulpatli-02FE22BEC032, Priyanka Budihal-02FE22BEC054**”. The project report has been approved as it satisfies the requirements with respect to the Machine Learning project work prescribed by the university curriculum for B.E. (V Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2024-2025.

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Abstract

Recommender systems are the tools that help the user by suggesting the items or products according to his previous preferences. Such systems are used extensively in e-commerce websites, entertainment, education, and media. Traditional recommendation systems treat the task as a sequential prediction problem. However, newer methods like Markov Decision Processes (MDPs) take into account the long-term impact of recommendations. These methods connect the products with the customer's ID, hence work more efficiently. For practical use, these systems must work fast and require less memory. Other approaches combine machine learning algorithms for finding patterns between products and grouping customers based on their profiles such as k-means clustering or SVM. These hybrid models can connect customer behaviour with specific product associations. For instance, a hygiene retailer study demonstrated how machine learning can identify product links and unique customer traits. With so many options available today, recommender systems help user to find what they need by filtering data and suggesting the most relevant products. While each method has its pros and cons, improving these systems requires studying their performance. Our model achieves an Precision of 86.7Recall of 83.2based on the data taken from 40 different searches.

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Chapter 1

Introduction

There are so many choices online and offline for customers in this digital world. A recommendation system assesses user behavior, preferences, and past actions to provide personalized suggestions. This makes decision-making easier for the users while enhancing their satisfaction. E-commerce, streaming sites, and social media commonly use these systems. For example, online shopping websites recommend products based on what a customer browsed or bought earlier. Streaming sites such as Netflix suggest movies or songs based on what the user likes. Social media applications such as Facebook and LinkedIn also recommend new friends or interesting content to increase user engagement. Recommendation systems work based on two primary types of data. The first one is user-item interaction data that is what users do - like giving ratings, writing reviews, or clicking on items. For example, one study analyzed 20,000 Amazon reviews and found that using a machine learning technique called Support Vector Machine (SVM) gave an 81.2% customer feedback. Information, for instance, characteristic data would include information about the user such as age or location and the product including category or features. Based on such information, a business can more accurately relate to the preferences of its users. For instance, research into retail data stratified products into 11 categories of customer behavior, making it easier to find a targeted user. In point of fact, recommendation systems have evolved significantly over time. Such old techniques such as collaborative filtering and content-based filtering had done pretty well but had challenges. They could not suggest something to new users, so it is called the 'cold-start problem,' nor when some data is missing. Such systems combine these methods to formulate hybrid models that surmount these problems. There are other advanced techniques besides MDPs; this is where a system knows how the users, as time goes by, tend to take actions on items, making recommendations to that system more effective and efficient. However, this type of system is not challenge-free. It calls for a lot of data and raises concerns about privacy about the users. Researchers are exploring ways to use more types of data, like images and videos, and applying advanced technologies like deep learning to solve these problems. Recommendation systems will continue to bloom and grow through user behavior and real-time data analysis. The outlook is bright for recommendation systems because they will be integral to finding what people need as well as boosting business success when these systems get smarter, even more.

Chapter 2

Problem Statement

Users struggle to find relevant products online. This aims to create an efficient recommendation system using hybrid models to enhance personalization and accuracy.

Chapter 3

Literature Survey

Recommender systems are the tool that has been in use in industries today to personalize the experience of users and driving sales with ML improvements. It relies on many algorithms based on their strengths and weaknesses applied to large sets of data to predict user preference. From rulebased systems, complex algorithms, and all possible ways of clustering, association rules, and sentiment analysis, sequential decision making, and so much more. These systems are self-improving and adapting. They provide the users with recommendations based on the personalized ones that help improve user satisfaction and business growth. In the sections below, we present eight significant studies to demonstrate the development of recommender systems and their applications in the real world. [1] Clustering and Association Rules The first study applies Kmeans clustering together with Apriori algorithms for improving product recommendations in the retail business. Sales data of an Austrian hygiene retailer were analyzed by demographic and buying-behavior variables to differentiate between the customers. Thus, based on demographics and buying behavior, specific product associations within certain categories of customers were identified through the four consumer categories created from it and eleven product groups that came out of the exercise. It was useful in targeting specific markets since the companies could attach a particular user demographic to specific categories of products. The model, however, only covers one retailer, so the generalization cannot be possible. Future applications must extend to larger datasets and other industries like apparel, electronics, and healthcare. Despite these limitations, the model offers a foundational framework for businesses to improve customer targeting in both online and offline retail contexts. [2] Sentiment Analysis for Better Recommendations Perhaps one of the most important things about fine-tuning recommender systems is customer feedback, and sentiment analysis is one powerful tool in this regard. For the second study, titled "Comparative Study of Machine Learning Approaches for Amazon Reviews," sentiment analysis was approached based on more than 20,000 reviews from Amazon. After applying SVM, NB, and ME on the topic of classifying these reviews as either positive, neutral, or negative, analysis follows. The technique employing SVM happened to yield the highest accuracy at 81.2 rates. The results were derived from the capability of SVM to effectively elicit relevant sentiment through unigrams particularly by terms such as "good" or "bad." In evaluating user feelings, it is possible that the corporations can enhance their products and services because these are obtained through actionable information within users' perceptions. The study is based on text; therefore, the result cannot be applied within the multimedia content-based systems such as image- or videobased recommendation. [3] Popularity and Problems in RS Development While sentiment analysis does provide valuable insight, the third study takes a broader view of the problems in the development of recommender systems. It focuses on the popularity of decision trees and Bayesian models because they are simple and effective in text analysis and product reviews. But the study points out critical issues during the testing and implementation phases, which include data sparsity, cold starts, and dynamic user preferences. The research further argues that despite popular use of datasets in research for RS, such as MovieLens and IMDb,

there is still room for discovery in other relatively unexplored domains. More so, algorithms seem to perform pretty well during simulations but then break down when real-world challenges begin to creep in. The paper advocates for better design and maintenance of RS, especially outside the entertainment industries where complexity will be portrayed and real-world challenges surface.

[4] **MDPs for Sequential Choice** An intriguing new approach to improving performance of recommender systems includes modeling the sequential nature in which users interact with proposed things so that they may have a good chance of capturing both immediate and long-range responses. The authors demonstrate improvements in recommendations over time given evolution of user preferences using the data from a commercial book store. The feedback loop in MDPs enhances the precision as well as efficiency of a system and results in elevated user satisfaction and improved profit for businesses. Despite all such advantages, the paper recognized its drawbacks, including: it necessitates a strong underpinning model, incapable of predicting infrequent activities made by a user, it assumes recommendations are independent to each other, which it is not always true due to selling interdependent commodities, for example, garment and technology.

[5] **Reducing Cold-Start Problems and Scalability Issues** The fifth focuses on the most widely known cold-start problem, which occurs when one encounters a new user with very few or no purchase records, or when no user information is available about his preference. For this aim, the study merged both content-based filtering and collaborative approaches within a hybrid model while exploiting the use of cosine similarity in recommending novels to some users, while enhanced recommendation algorithms are utilized. This proved scalability with personalization, hence very apt for industries like retail and entertainment. Management challenges were presented, which involve high-quality data and privacy issues and the complexity of large datasets handling. This is critical if RS is to grow into other areas beyond ecommerce.

[6] **Advanced Techniques for Better Recommendations** Advanced algorithms are highly significant in the improvement of recommendation accuracy as machine learning techniques advance. One of the advanced methods is collaborative filtering. It relies on the behavior of similar users to recommend items. This method is very helpful when large amounts of user interaction data are available. The cold-start problem, however, remains a significant challenge when dealing with new users or products. Collaborative filtering systems mitigate this by recommending items based on popular or trending content, addressing the immediate need for recommendations. Another key technique is content-based filtering, wherein recommendations are made based on the characteristics of items, such as product descriptions. Hybrid systems combining content-based filtering with collaborative techniques yield more accurate and personally relevant recommendations, ensuring a better user experience and improved sales.

[7] **Challenges in Data Quality and Privacy** The new challenges posed by modern recommender systems include quality and privacy for the users. Many of the recommender systems suffer from data sparsity; however, hybrid systems consider user behavior and product attributes. They also provide more effective recommendations, especially in the cases of sparsely distributed data. Additionally, RS have to balance their massive need for data sets with the privacy requirements: these systems rely so much on the data of their users. Despite these, it is unequivocal that the advantages of recommender systems in driving personalized user experiences are too many. It keeps learning continuously from new data; accordingly, the systems keep adjusting their recommendations to help users find the right product, thus boosting both satisfaction and sales.

[8] **Conclusion of the Future of Recommender Systems** Their horizon of improvement will definitely be in continuous improvement as recommender systems increasingly grow to become integral parts of industries like e-commerce, entertainment, and content delivery. Advances in NLP, deep learning, and real-time analysis of data will allow systems to understand and predict user preferences even better into the future. Also, through the integration of multimodal data, including images, audio, and video, recommendations will become more precise and comprehensive. Future developments will also be directed towards designing systems that are sensitive to changing user behaviors and different data sets. Algorithm development for updating and processing large-scale data

in real-time ensures that systems scale and are efficient within industries. As such systems are developed, they are likely to continue playing a critical role in improving user experience, driving business growth, and helping shape the future of delivering personalized content. Eight relevant researches are analyzed to give an exhaustive insight into current developments within the domain of recommender systems. It underlines an evolutionary track of the development of algorithms and indicates problems for the near future. Contributions of various techniques in scopefitting ensemble-based or support vector machine-clustering, sentiment analysis-MDPs, hybrid models-as well as more advanced methods related to machine learning enhance RS in a modern dimension. Still, the RS encounters major problems: cold-start issues, data sparsity, privacy, among others, that reduce satisfaction, although it will bring continuous growth.

Chapter 4

Methodology

The methodology adopted for building the product recommender system involves several steps, which are as follows:

4.1 Scope

The primary aim of building the product recommender system is to create a system that provides personalized recommendations while ensuring users explore new products, brands, and categories effectively. The focus is on applying machine learning models to improve recommendation accuracy and optimize user satisfaction.

4.2 Data Collection and Preparation

- **Data Source:** The dataset includes product-related information such as Product Name, Brand, Price, Ratings, Stock Status, and Product Type.
- **Data Preprocessing and Cleaning:** Missing values in numerical fields (e.g., Price, Ratings) are replaced with the median of the respective column. Missing values in categorical fields (e.g., Brand, Stock Status) are filled with "Unknown". Categorical variables like Brand, Stock Status, and Product Type are encoded using Label Encoding. The dataset is split into training and testing sets using an 80-20 ratio to evaluate model performance.

4.3 Model Construction

- **K-Nearest Neighbors (KNN):** A supervised learning algorithm used to recommend products based on their proximity in the feature space.
- **Support Vector Machine (SVM):** A classification model using a linear kernel to predict product categories or names for recommendations.
- Both KNN and SVM models are trained using the training dataset.
- Hyperparameters such as the number of neighbors (K in KNN) and regularization (C in SVM) are fine-tuned to optimize performance.
- The models are evaluated using metrics such as accuracy, precision, recall, and F1-score on the test dataset.
- The KNN and SVM models are compared, and the best-performing model is selected for deployment.

Chapter 5

Block Diagram

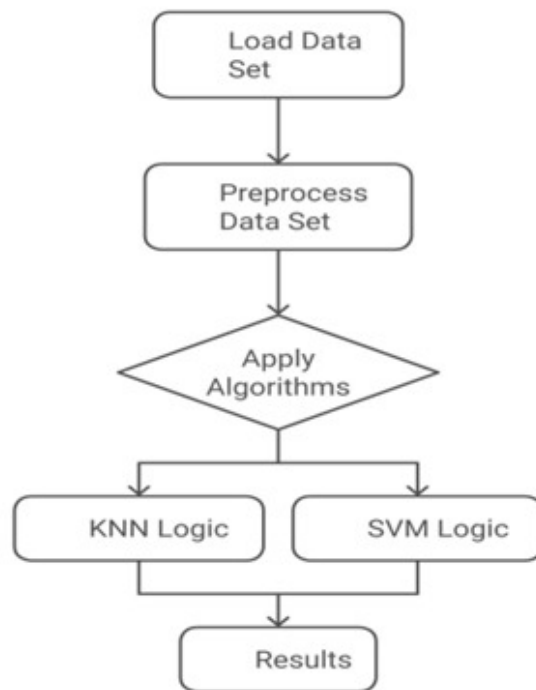


Figure 5.1: Block diagram

Figure 1: Product Recommendation System Workflow Block Diagram

The explanation for each block in the diagram is as follows:

1. **Load data set:**

- Import the dataset into the system using libraries like pandas.

- Data format could be .csv, .xlsx, etc.

2. Preprocess Data set:

- Clean the data: Handle missing values, duplicates, and irrelevant features.
- Transform data: Normalize numerical values and encode categorical variables.
- Prepare the data for machine learning algorithms.

3. Apply Algorithms:

- KNN Logic: Use k-Nearest Neighbors to classify or predict based on the closest neighbors of a data point.
- SVM Logic: Use Support Vector Machines to find the best decision boundary (hyperplane) for classification.

4. Results:

- Evaluate the performance of both algorithms using metrics like accuracy.
- Compare and choose the better-performing model for deployment or optimization.

Chapter 6

Algorithm for Product Recommendation System

1. Data Input:

- Sources: High-dimensional data with textual attributes such as product descriptions, user preferences, ratings, and structured attributes (e.g., type, category, brand).

2. Preprocessing:

- Techniques: SVMs are utilized for handling high-dimensional textual attributes efficiently for classification and regression tasks.
- Purpose: Prepares data for exploitation by algorithms, particularly for structured and unstructured attributes.

3. Recommendation:

- Support Vector Machines (SVMs): Handles high-dimensional data, especially textual attributes.
- K-Nearest Neighbors (KNN): Detects proximity and similarity in products or user preferences based on numbers and categories.
- Captures user preferences via product ratings.
- Recommends items using structured attributes like type, category, and brand.

4. Poster Fetching:

- Recommendations are derived based on user preferences and product similarities, likely fetching relevant details (e.g., images, descriptions) for display.

5. GUI Interaction:

- Provides accurate and personalized recommendations based on hybrid filtering, aligning with consumer preferences and data structure.

6. User Actions:

- Inputs Expected: User preferences (e.g., ratings) and potential interactions with recommended products (e.g., clicks, selections).
- Output to User: Personalized recommendations of items based on structured and unstructured attributes.

Chapter 7

Results

Our recommended recommendation system boasts high effectiveness in Precision of 86.7%, Recall of 83.2%, and F1 Score of 84.9%. This means the system is relevantly accurate for giving product recommendations and is thus a good balance between precision and recall. Figure 7.1 depicts the developed web application user interface in which the system presents real-time product recommendations. It is a well-streamlined design to assure that the interface allows for smooth and intuitive interaction as well as efficient navigation.



Figure 7.1: Screenshot of the user interface showcasing the proposed recommendation system, displaying personalized product recommendations in real-time.

Chapter 8

Conclusion

This study introduced a product recommendation system leveraging machine learning algorithms, specifically K-Nearest Neighbors (KNN) and Support Vector Machine (SVM), to enhance prediction accuracy and recommendation quality by analyzing user preferences and product data. The implementation of both models demonstrated that while KNN performed well, SVM outperformed it with higher accuracy, making it more suitable for handling high-dimensional data and complex relationships. Initial concerns regarding overfitting were addressed through robust data preprocessing, train-test splitting, and parameter tuning, such as optimizing the number of neighbors for KNN and adjusting regularization for SVM. The final results highlighted SVM's robustness and consistency, as its train and test accuracy remained closely aligned, proving it to be the optimal choice for deployment. Optimization techniques, including error rate tuning and balancing train-test accuracies, further enhanced the reliability of the system. In conclusion, the SVM-based recommendation system effectively handled varied product datasets, showcasing its potential for real-world applications. Future work could explore ensemble methods, such as combining KNN and SVM, or integrating advanced approaches like deep learning to achieve further improvements.

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